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Correct Answer

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Practice Exam - 1

Question 67 of 117



The patient is a 45-year-old man with a PMH of HTN, obesity, diabetes mellitus and paroxysmal atrial fibrillation presenting for evaluation of slowly progressive severe dyspnea on exertion for 1 year. He has had many presentations with rapid atrial fibrillation and to date has been treated with multiple anti-arrhythmics and 4 atrial fibrillation ablations. At this time, recent ambulatory event monitors suggest no breakthrough atrial fibrillation within the last year. Surface echo showed a preserved EF, normal wall thickness, biatrial enlargement, and grade II diastolic dysfunction without significant resting valvular abnormalities. A myocardial spect is normal. A cardiac CTA and cardiac pulmonary vein CT are normal. A VQ scan is unremarkable. Moderate dose diuretics are offered without relief and cause orthostasis. He underwent an exercise left and right heart catheterization which included a transseptal puncture to measure left atrial pressure directly. The values from this study are below:

	HR	BP	RAP	mPAP	LAP	LVEDP	V wave
Rest	73	120/80	2	18	9	7	16
Moderate exercise (30 w)	86	140/70	19	50	35	15	44
Peak exercise (70 w)	110	160/55	21	60	38	17	48

What is the most likely etiology of this patient's symptoms?

A. Dynamic mitral valve stenosis

B. Pulmonary vein stenosis

C. Stiff left atrial syndrome

D. idiopathic heart failure with preserved ejection fraction

E. cardiac amyloidosis

F. idiopathic exercise induced pulmonary hypertension

Commentary

References

Testing point: The Board-certified AHFTC specialist should recognize etiologies of dyspnea on exertion that may mimic heart failure with preserved ejection fraction but have different prognoses and treatment strategies.

Answer: C. stiff left atrial syndrome.

The patient presents with worsening dyspnea on exertion. He carries a history of pAF and multiple ablations. His AF has been reasonably controlled. His extensive cardiovascular workup excludes many common etiologies of dyspnea including coronary disease, valvular heart disease, and PE. The exercise study shows normal resting hemodynamics. His blood pressure and heart rate response are relatively unremarkable. With exercise, there is a minimal elevation in his LVEDP but a dramatic elevation in his left atrial pressure. In the absence of mitral regurgitation, this likely represents an atrial compliance abnormality.

Stiff left atrial syndrome has been well described in the literature as a complication of repeated ablation that results in extensive left atrial scarring. He does not respond to diuretics because at rest he is not generally volume overloaded and his hemodynamic abnormality is largely with exertion. There is no specific treatment for stiff left atrial syndrome, though transplant may be definitive and atrial septostomy could be considered. It is however important to diagnose and differentiate this disorder from similarly presenting forms of exertional dyspnea such as pulmonary vein stenosis, HFpEF, cardiac amyloidosis, and idiopathic exercise induced pulmonary hypertension.

Answer A: Dynamic mitral valve regurgitation can cause similar hemodynamics and should be sought in such a case, but dynamic mitral valve stenosis, answer A, is not a known entity.

Answer B: No pulmonary vein stenosis is seen on his chest CT.

Answer D: Idiopathic HFpEF is a diagnosis of exclusion. It can present with similar hemodynamics, but the context of this case makes C a better choice.

Answer E: His echo is not suggestive of amyloidosis.

Answer F: A CTA pulmonary vein does not find pulmonary stenosis. The patient does have pulmonary hypertension during exercise, this is secondary to stiff left atrial syndrome and not idiopathic in nature.

In addition this case illustrates nicely the interventricular interaction that can be seen in cases of HFpEF, stiff left atrial syndrome and mitral valve disease, where a low left ventricular transmural pressure, defined as the difference between the left ventricular end diastolic and right atrial pressure, suggests right ventricle pressure overload during exercise contributed to the small increases in LV filling pressures seen. His right sided filling pressures dramatically increase as a result of the rise in his left atrial pressures. Large V waves are generated as well, again a sign of left atrial non-compliance.

